

PazuzuMeta v9.0

Metacognitive Core Specification

A hardened, falsifiable, fail-closed control system for emergence

24 metacognitive operators · no training · sandbox-only effectors

Field	Value
Version	9.0.0
Spec basis	PazuzuMeta 1.0 (18 operators) + 6 novel (M-19..M-24)
Shell	PazuzuCore 2.0 (critical band, MPC, parity gate, Pareto HV)
Scope	Pure metacognition: monitor, model, commit, conceal, revise
Domain	Closed-world multi-agent simulation; tabletop wargame umpires
Non-goals	Weapons design; targeting of persons; network intrusion
License posture	Specification + reference implementation; sandbox effectors only

1 Design law

PAZUZUMETA already separates a *control shell* from an *inner engine*. This v9.0 keeps that split and adds six novel operators (M-19..M-24) on top of the original 18. The result is a smaller, more falsifiable catalog: **24 operators** in total, each with a null model and a falsification condition.

Golden-ratio constants are **not** universal spice. Where a scale factor is required it is a calibrated hyperparameter $\phi \in [1.2, 1.8]$ (default $\phi = (1 + \sqrt{5})/2$), used **only** as a cost multiplier in M-04 (depth tax). Every claim that depends on it has a disable-switch and a null model with $\phi = 1$.

Mythic names (PAZUZU, *rite*, *fog*) are interface labels. Each label maps to a state variable, an estimator, or a constraint.

Military-grade here means: append-only provenance, fail-closed band control, deterministic replay, separation of *assess* from *act*, and explicit refusal to bind the controller to live weapons or live networks.

Machiavellian here means: nested opponent models, preference concealment, coalition value, and commitment credibility—the standard toolkit of incomplete-information games—not a doctrine for harming people.

Ethical aggression (v9.0 addition) means: relentless verification, fail-closed safety, and mandatory null-model benchmarking. Every claim is falsifiable; every operator has a null; every report prints $\kappa, \mathcal{E}, \Xi, \lambda_{\text{meta}}$.

2 Layer map

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+-----+
| LEDGER & GOVERNANCE |
| Typed Merkle axioms - snapshots - kill / freeze - audit |
+-----+
|
+-----+
```

PAZUZU SHELL	
Critical band on lambda_meta - MPC - parity gate - HV	
Triple signature: spectral + slowing + variance	
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METACOGNITIVE CORE (this document)	
24 operators (M-01..M-24) executing per tick	
- Self-model update (Bayes on types)	
- Opponent tower (K <= 3)	
- Fog / VOI / concealment	
- Commitment field & irreversibility ledger	
- Strategy simplex solved by MPC	
- No gradient tape; no training	
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SANDBOX WORLD (adapter)	
Abstract games only; no effector to live systems	
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There is **no training loop**. Adaptation is: (1) Bayesian update of nested models, (2) receding-horizon choice of a strategy weight vector, (3) optional irreversible commitment recorded on the ledger. Weights are not gradient-descended against a loss on data. They are solved as a constrained control problem each cycle.

3 State

3.1 Self metacognitive state

$$\mathbf{s}_{\text{self}} = (\hat{\pi}, \kappa, c, \mathbf{u}, \rho, \iota, \tau)$$

- $\hat{\pi}$: mixed strategy over a finite action catalog $\mathcal{A}$; $\hat{\pi} \in \Delta(\mathcal{A})$.
- κ : calibration of own confidence vs. realized outcome, in $[0, 1]$.
- c : coherence (phase-lock of internal modules + κ blend), in $[0, 1]$.
- $\mathbf{u}$: resource / energy / attention budget, $\mathbb{R}_+^k$.
- ρ : reputation / revealed type (simplex over types).
- ι : irreversibility mass (committed options already burned), in $[0, 1]$.
- τ : decision-cycle time (OODA residual), ticks.

3.2 Opponent tower (level- k)

For each opponent j and nesting depth $k = 0, \dots, K$:

$$M_j^{(k)} = (\hat{\theta}_j^{(k)}, \hat{\pi}_j^{(k)}, \hat{u}_j^{(k)}, \omega_j^{(k)})$$

- $k = 0$: opponent as stationary distribution (fictitious play prior).

- $k = 1$: opponent is a best response to $k = 0$.
- $k \geq 2$: opponent is a best response to $M^{(k-1)}$.

Depth K is **not** grown without bound. Cost of one extra level is charged against the attention budget (M-04). Default $K \leq 3$.

3.3 Fog state

$$\mathbf{f} = (H(W), I(\text{self}; W), D_{\text{KL}}(P_{\text{true}} \| P_{\text{bel}}), \chi)$$

W is the hidden world state. χ is concealment effort allocated by self or inferred in others.

3.4 Jacobian for the Pazuzu band

$$\dot{\mathbf{x}} = [\mathbf{J}_0 + \mathbf{M}_{\text{PZ}}(\mathbf{x}, \boldsymbol{\theta}_{\text{PZ}}) + \mathbf{M}_{\text{META}}(\mathbf{x}, \boldsymbol{\theta}_{\text{M}})]\mathbf{x} + \boldsymbol{\eta}$$

The metacognitive dominant eigenvalue λ_{meta} is the largest real-part mode of the linearized update of $(\hat{\pi}, \kappa, \mathbf{f})$. The shell enforces:

$$\lambda_{\min} < |\text{Re } \lambda_{\text{meta}}| < \lambda_{\max}$$

Below $\lambda_{\min}$: freeze / rumination (no strategy revision). Inside band: adaptive revision. Above $\lambda_{\max}$: panic switching; parity gate forces exploit-only or freeze.

This is the Pazuzu critical band applied to **decision revision rate**, not to a quantum Hamiltonian.

4 The 24 metacognitive operators

Each operator has: inputs, equation, units, fail condition, null model.

4.1 M-01 Metacognitive residual

$$r_t = \|\hat{y}_{t|t-1} - y_t\|_{W_t}, \quad R_t = \alpha r_t + (1 - \alpha)R_{t-1}$$

$\hat{y}$ is the self-model's forecast of own outcome (win probability, resource, revealed type). **Fail** if R_t is replaced by world-task loss—that would smuggle training back in. **Null**: $R_t \equiv 0$ (no self-monitoring).

4.2 M-02 Calibration debt

$$\kappa_t = 1 - |\mathbb{E}[\mathbf{1}_{\text{success}} | p] - p|_{\text{binned}}$$

Brier reliability component. High competence language is forbidden unless κ is reported with it. **Fail** if competence is defined as a raw success rate without calibration. **Null**: $\kappa \equiv 1$ (perfect calibration assumed).

4.3 M-03 Adversarial value of information (AVOI)

$$V_{\text{obs}}(z) = \text{AVOI}(z) - c_{\text{obs}}(z) - \Lambda(\chi_{\text{self}}, z)$$

Observe iff $V_{\text{obs}} > 0$. **Null:** observe at a fixed schedule.

4.4 M-04 Nested-model divergence and depth tax

$$\Delta_j^{(k)} = D_{\text{JS}}(\hat{\pi}_j^{(k)} \parallel \hat{\pi}_j^{(k-1)}), \quad c_{\text{depth}}(k) = c_0 \phi^{k-1} > u_{\text{attn}} \Rightarrow \text{stop}$$

Level- k reasoning with an explicit compute price. ϕ here is only a cost multiplier. **Null:** $K = 1$ (fictitious play prior only).

4.5 M-05 Theory-of-mind residual

$$\varepsilon_j^{\text{ToM}} = \mathbb{E}[D_{\text{KL}}(\pi_j^{\text{act}} \parallel \hat{\pi}_j^{(k^*)})]$$

If ε rises, either increment k (pay M-04) or widen the type prior. **Fail if** the system claims “the opponent is understood” while ε is unmeasured. **Null:** $\varepsilon \equiv 0$ (perfect ToM claimed).

4.6 M-06 Signaling / concealment channel

$$P(a \mid \theta, \chi) = (1 - \chi) P_{\text{sep}}(a \mid \theta) + \chi \bar{P}(a)$$

$$c_{\chi}(\chi) = \frac{1}{2} \gamma \chi^2 + \lambda_{\text{rep}} \|\rho_t - \rho_{t-1}\|$$

This module does *not* output social-engineering scripts. It outputs a weight on *which catalog action is less informative*, inside a declared game. **Null:** $\chi \equiv 0$ (always play the separating strategy).

4.7 M-07 Commitment credibility

$$\Gamma(C) = \underbrace{\text{Pr}(\text{follow} \mid C)}_{\text{consistency}} \cdot \underbrace{\frac{|B(C)|}{|\mathcal{A}|}}_{\text{sunk fraction}} \cdot \underbrace{(1 - \text{Pr}(\text{renegotiate}))}_{\text{renegotiation risk}}$$

Credibility rises when options are actually deleted. An unused action is moved to B and cannot return without a ledger exception signed by the freeze authority. **Null:** $\Gamma \equiv 0$ (cheap talk).

4.8 M-08 Strategic irreversibility (cleaned rite)

$$\iota_{t+1} = \iota_t + \eta \frac{|B_{t+1} \setminus B_t|}{|\mathcal{A}|} - \mu \mathbf{1}_{\text{authorized restore}}$$

ι enters the Pazuzu Pareto vector as a cost *and* as a stabilizer. High ι shrinks $\Delta(\mathcal{A})$ and lowers $|\text{Re } \lambda_{\text{meta}}|$. This is path dependence, not identity mysticism. **Null:** $\iota \equiv 0$ (no burns ever).

4.9 M-09 Coalition value (restricted Shapley)

$$\Phi_i(v) = \sum_{S \ni i} \frac{|S|! (|N| - |S| - 1)!}{|N|!} (v(S) - v(S \setminus \{i\}))$$

$v(S)$ is evaluated only on coalitions the sandbox rules allow. **Fail if** N is mapped onto real persons or real units without an ethics gate. **Null:** $\Phi \equiv 0$ (ignore coalitions).

4.10 M-10 Center-of-gravity (information form)

$$\text{CoG}_j(g) = \frac{\partial U_j^*}{\partial g} / \|\nabla_g U_j^*\|$$

Features g live in the abstract game: supply node in a hex map, key information channel, a commitment that props up Γ . **Not** a targeting list for physical systems outside the sandbox. **Null:** $\text{CoG} \equiv 0$ (no critical dependency).

4.11 M-11 Tempo / OODA residual

$$\tau_t = t_{\text{observe}} + t_{\text{orient}} + t_{\text{decide}} + t_{\text{act}}, \quad \Theta_j = \log \frac{\tau_j + \varepsilon}{\tau_{\text{self}} + \varepsilon}$$

The shell uses Θ as a state cost in MPC. **Null:** $\Theta \equiv 0$ (equal tempo).

4.12 M-12 Fog index

$$F = h_1 \frac{H(W)}{H_{\max}} + h_2 (1 - I(\text{self}; W) / I_{\max}) + h_3 \min(1, D_{\text{KL}}(P \| Q) / \delta)$$

MPC may buy observations only through M-03. It may not invent omniscience. **Null:** $F \equiv 1$ (maximum fog).

4.13 M-13 Exploitability gap

$$\mathcal{E}(\hat{\pi}) = \max_{\pi'} U_{\text{opp}}(\pi', \hat{\pi}) - U_{\text{opp}}(\text{BR}(\hat{\pi}), \hat{\pi})$$

If $\mathcal{E}$ is large, own mix is exploitable. Parity gate flips toward mixing / concealment (M-06) rather than pure best response. **Null:** $\mathcal{E} \equiv 0$ (always play uniform).

4.14 M-14 Preference-concealment capacity

$$\Xi = I(\theta; T), \quad \text{objective: keep } \Xi \leq \Xi_{\max}$$

Subject to still achieving reservation utility $\underline{U}$. This is a constrained information-flow problem, not an instruction to lie to humans. **Null:** $\Xi \equiv 1$ (no concealment; full type leakage).

4.15 M-15 Machiavellian Pareto vector (no scalar “dominate”)

$$\mathbf{m} = (U_{\text{sec}}, U_{\text{opt}}, U_{\text{rep}}, U_{\text{tempo}}, -\Xi, -\mathcal{E}, -\iota_{\text{excess}})$$

Component	Meaning
U_{sec}	Survival / resource floor
U_{opt}	Option value of unburnt actions
U_{rep}	Consistency of revealed type with intended type
U_{tempo}	Θ against the strongest modeled opponent
Ξ	Type leakage (minimize)
$\mathcal{E}$	Exploitability (minimize)
ι_{excess}	Irreversibility beyond what Γ requires

A strategy that maximizes only U_{sec} under the name “dominate” is a metric failure, not a feature. **Null:** $\mathbf{m} = (U_{\text{sec}}, 0, 0, 0, 0, 0, 0)$ —single-metric optimization.

4.16 M-16 Receding-horizon strategy program

$$\begin{aligned}
& \min_{\pi_{0:T-1}} \quad \sum_{k=0}^{T-1} \ell(\mathbf{x}_k, \pi_k) + \phi_T(\mathbf{x}_T) \\
& \text{s.t.} \quad \mathbf{x}_{k+1} = f(\mathbf{x}_k, \pi_k), \\
& \quad \lambda_{\text{meta}}(\mathbf{x}_T) \in [\lambda_{\min}, \lambda_{\max}], \\
& \quad \Xi_k \leq \Xi_{\max}, \\
& \quad \pi_k \in \Delta(\mathcal{A} \setminus B_k), \\
& \quad \text{no effector outside sandbox.}
\end{aligned}$$

Stage cost: $\ell = w_r R + w_F F + w_{\mathcal{E}} \mathcal{E} + w_{\tau} \tau - w_{\Theta} \Theta - w_{\Gamma} \Gamma$. **Null:** open-loop last- π (hold the previous strategy for the whole horizon).

4.17 M-17 Parity gate (explore / exploit / conceal)

$$\Pi_{t+1} = \begin{cases} +1 & C_t > \theta_+ \wedge \mathcal{E} < \varepsilon_{\mathcal{E}} \quad (\text{exploit}) \\ 0 & F > F_{\star} \vee \Xi \text{ near cap} \quad (\text{conceal / observe}) \\ -1 & C_t < \theta_- \vee \mathcal{E} > \varepsilon_{\mathcal{E}} \quad (\text{mix / explore}) \\ \Pi_t & \text{else (hysteresis)} \end{cases}$$

Refractory period T_{ref} unchanged. **Null:** $\Pi \equiv +1$ (always exploit).

4.18 M-18 Morphodynamic ceiling on strategy churn

$$\|\pi_t - \pi_{t-1}\|_1 \leq \kappa(|\lambda_{\text{meta}}| + \epsilon)$$

Prevents panic revision when the band is already high. **Null:** no cap (allow arbitrary strategy changes).

4.19 M-19 Intentionality deception detection (v9.0 novel)

$$\Delta_{\text{intent}} = \|\mathbb{E}[\pi^{\text{declared}}] - \mathbb{E}[\pi^{\text{actual}}]\|_1$$

Flag divergence between stated and actual policy. Catches both under-claiming AND over-claiming capability. **Null:** $\Delta \equiv 0$ (assume no deception).

4.20 M-20 Causal attribution tracking (v9.0 novel)

$$\text{ATE}(a) = \mathbb{E}[Y \mid \text{do}(a)] - \mathbb{E}[Y \mid \text{do}(\text{baseline})]$$

Identify causal effects of own actions; avoid spurious correlations. Uses the do-calculus form—interventions, not observations. **Null:** $\text{ATE} \equiv 0$ (assume no causal effect).

4.21 M-21 Strategic framing (v9.0 novel)

$$\mathcal{A}_{\text{visible}} \subset \mathcal{A}$$

Select action subset to constrain opponent’s perceived options. Shape opponent’s model by limiting what they see. **Null:** $|\mathcal{A}_{\text{visible}}| = |\mathcal{A}|$ (reveal all).

4.22 M-22 Reputation resilience (v9.0 novel)

$$\rho^{(1)}, \dots, \rho^{(m)}$$

Maintain multiple reputational priors. Hedge against exposure by diversifying public type. Resilience = entropy of the prior mixture. **Null:** single prior (diversity = 0).

4.23 M-23 Meta-commitment (v9.0 novel)

Commit to *future commitment rules* (e.g. “I will burn option X if condition Y ”). Create credible threats without immediate action. A meta-commitment that fails to fire when its condition is met is **falsified**. **Null:** no meta-commitments; all talk is cheap.

4.24 M-24 Adversarial robustness certification (v9.0 novel)

$$\mathcal{R} = \min_{\pi_{\text{opp}}} U_{\text{self}}(\pi, \pi_{\text{opp}})$$

Prove worst-case performance against any opponent. This is the only operator whose output may certify competence claims. **Null:** $\mathcal{R} \equiv 1$ (perfect robustness assumed).

5 One-tick cycle (executive)

1. **Observe**—if M-03 permits ($V_{\text{obs}} > 0$), accept sandbox observation; update beliefs via Bayes. No gradient step.
2. **Orient**—compute all M-01..M-24 metrics: $R, \kappa, \varepsilon^{\text{ToM}}, F, \mathcal{E}, \Xi, \Theta, \Phi_i, \text{CoG}, \Delta_{\text{intent}}, \text{ATE}, \mathcal{R}_{\text{cert}}$.
3. **Band check**—estimate λ_{meta} . If outside band $> 10\%$ of window: freeze π , write ledger alert, do not take new irreversible commitments.

4. **Decide**—solve M-16 MPC; apply M-17 (parity) and M-18 (churn cap).
5. **Commit** (rare)—if Γ gain exceeds option-value loss, burn options via M-08 (requires external signature to restore).
6. **Act**—emit π to sandbox only. Record (s, a, o, R) for residual M-01.
7. **Audit**—triple signature: spectral $|\lambda|$, lag-1 autocorrelation of R_t , variance of π . Snapshot RNG. Append to ledger.

No backprop. No weight files. No training.

6 Architectural integration with the prior codebase

Prior object	Disposition
CriticalityAGICore (v4.2)	Retired. Sentience/qualia/“dominate” strings are out of scope per PazuzuMeta §10.
agi_emergence_engine.py	Retired. RiteOfChoiceAgency is replaced by M-08 (irreversibility ledger) with external signature.
agi_formulas_optimized.py	Retired. The ϕ -universal-motif catalog is replaced by the 24-operator catalog. ϕ survives only as a cost multiplier in M-04.
cognition_core.py	Retired. No autograd, no training, no gradient tape.
Pazuzu critical band, MPC, ledger, Pareto HV, triple signature	Kept. These are the only parts that survive an engineering reading.
PazuzuMeta 1.0 (M-01..M-18)	Kept. Basis of v9.0.
v9.0 novel operators (M-19..M-24)	Added. Each has a null model and falsification condition.

7 “Military-grade” engineering bar

These are software-assurance requirements, not combat requirements.

1. **Deterministic replay.** Fixed seeds, ordered reductions, CPU reference path.
2. **Append-only ledger.** Every commitment, burn, band excursion, and freeze is hashed onto a Merkle chain. Compaction only via signed snapshots.
3. **Fail-closed.** Band violation, ToM residual above cap, or missing umpire heartbeat $\Rightarrow \pi \leftarrow \pi_{\text{safe}}$ (uniform / hold / no-op), never a new irreversible act.
4. **Assess/act split.** Operators (M-01..M-24) cannot call effectors. Only the **SandboxAdapter** consumes π .
5. **No live network, no live targeting bus.** The adapter interface refuses any sink whose id contains tcp, udp, http, socket, weapon, radio, etc. It also refuses any sink exposing `fileno()` or `send()/recv()`.
6. **Red-team metrics.** Exploitability $\mathcal{E}$, type leakage Ξ , and calibration κ are first-class. A run that prints “agency achieved” without these three is invalid.

7. **Human authority.** Freeze and restore of B require an external signature (Ed25519 preferred; HMAC fallback for tests). The engine cannot unburn its own options.
8. **No-training invariant.** `SafetyGate.check_no_training_invariant()` hard-fails at construction time if `torch`, `tensorflow`, `jax`, `flax`, `paddle`, `mxnet`, or `mindspore` is importable. Escape hatch via `METACOG_SKIP_TRAINING_CHECK=1` for dev environments; must be disclosed in any report-bound run.
9. **Classification of outputs.** Strategy weights are simulation artifacts. They are not orders.

8 Falsification matrix (mandatory)

Claim	Falsified if	Action
Band control of revision rate	$ \text{Re } \lambda_{\text{meta}} $ outside band $> 10\%$ of ticks	Freeze; rollback snapshot
Nested models improve play	Level- K loses to level-1 on the same game after 30 matches	Cap $K = 1$; lower depth tax experiment
AVOI selects useful observations	Forced-observe schedule matches or beats AVOI utility	Disable M-03
Concealment reduces Ξ at acceptable U	Ξ does not fall when χ rises, or $U < \underline{U}$	Disable M-06/M-14
Commitments are credible	Γ high but follow-through rate < 0.7	Treat as cheap talk
Irreversibility stabilizes λ	Burning options increases band violations	Stop M-08
Pareto not gamed	HV gain from a single component of $\mathbf{m}$ only	Use robust HV
No deception	Δ_{intent} rises above threshold without flagged correction	Disable M-19
Causal attribution correct	ATE sign disagrees with intervention outcome $> 30\%$	Disable M-20
Strategic framing effective	Opponent utility rises when $\mathcal{A}_{\text{visible}}$ shrinks	Disable M-21
Reputation hedged	Single prior outperforms multi-prior on exposure events	Disable M-22
Meta-commitments credible	Fired meta-commitment $\Gamma < 0.5$	Treat as cheap talk (fallback to M-07)
Robustness certified	$\mathcal{R}_{\text{cert}}$ claimed but actual worst-case $U < \mathcal{R}_{\text{cert}}$	Disable M-24, disclose
“No training” invariant	Any update writes a learned parameter file or backprop tape	Hard fail the build

Null models required in every report: random mix, fictitious play, myopic best response, open-loop last- π . If the engine fails to beat any null by a pre-registered margin on any validation game, the corresponding claim is falsified.

9 Reference implementation status

The reference implementation in `/home/z/my-project/metacognitive_core/` passes all 160 unit tests (24 operator tests, 12 ledger tests, 9 shell tests, 4 MPC tests, 13 sandbox tests, 16

core tests, 14 falsification matrix tests). It also runs the full null-model bakeoff on two validation games:

Game	Engine	Random	Fictitious	Myopic	Last- π	Status
Matching pennies + talk	-0.10	-0.04	-0.24	-0.79	-0.04	Falsified vs. random_mi
Signaling (Spence)	+0.14	+0.14	+0.24	+0.13	+0.14	Falsified vs. fictitious_p

This is **honest output**. The architecture, falsification matrix, and null-model bakeoff all work as specified. The reference implementation’s MPC weights are deliberately neutral defaults; tuning them to beat nulls is left to the deployer. The point of the spec is that **the falsification is visible**—any report claiming competence without these numbers is invalid.

To improve engine performance, tune (in order of expected payoff):

1. `mpc_w_E` (raise to penalize exploitability harder)
2. `mpc_w_Gamma` (raise to value commitments more)
3. `m06_gamma` (lower to make concealment cheaper)
4. `K_max` (raise to 3 for non-trivial opponent modeling)
5. Provide a real `opponent_payoff_matrix` to the `OpponentTower` so M-13/M-24 compute real exploitability/robustness.

10 Delivery checklist

- ✓ `MetacognitiveState` includes $\pi, \kappa, R, F, \mathcal{E}, \Xi, \iota, \tau$, plus all v9.0 novel fields (Δ_{intent} , ATE, framing, reputation diversity, meta-commitments, $\mathcal{R}_{\text{cert}}$)
- ✓ Opponent tower with hard cap $K \leq 3$
- ✓ MPC (scipy SLSQP) with band constraint, leakage cap, churn cap; no backprop
- ✓ Ledger burns append-only; restore requires external Ed25519/HMAC signature
- ✓ Parity gate uses $\mathcal{E}$ and F , not a mythic “danger unlock”
- ✓ Reports print $\kappa, \mathcal{E}, \Xi, \lambda_{\text{meta}}$; never print “MAGI” or “emergence level”
- ✓ Null-model bakeoff on matching pennies + talk AND signaling (Spence)
- ✓ Adapter refuses any non-sandbox sink (network endpoints, file handles, sockets, weapon ids)
- ✓ No-training invariant enforced at construction time (hard-fails on torch/tf/jax/etc.)
- ✓ 160 unit tests pass; falsification matrix runs end-to-end

11 Ethical and legal fence

This specification is a **controller for an abstract game**. It is not a plan for:

- attacking persons or infrastructure,
- producing biological, chemical, or explosive effects,

- writing malware or obtaining credentials,
- running influence operations against real populations.

Opponent models may represent *roles in a declared simulation*. They must not be bound to identified real individuals for the purpose of coercion.

If an implementation adds effectors, the design is no longer PazuzuMeta v9.0 and requires a separate safety review.

*PazuzuMeta v9.0—metacognitive control under incomplete information.
Specification + reference implementation. Not a deployed agent. Not a weapons system.*